



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 31

Level 2 — Intermediate · Fractions II

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Fractions II — Mixed Numbers & Improper Fractions
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	31 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 31 introduces mixed numbers and improper fractions — converting both ways, comparing $\frac{7}{4}$ with $1\frac{1}{2}$, placing fractions on the number line, and multi-step stories: milk drunk over a week, ropes joined, marbles given away in two different fractions. The bottle question at the end is a genuine three-step Olympiad-style finisher.

Learning Objectives

- Convert between mixed numbers and improper fractions both ways.
- Add and subtract with mixed numbers and wholes ($2 - \frac{3}{4}$).
- Place fractions between whole numbers on a number line.
- Simplify fractions and recognise equal mixed/improper pairs.
- Solve multi-step fraction stories with three changes.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. The mixed number **2 and $\frac{1}{3}$** written as an improper fraction is:

- (A) $\frac{5}{3}$ (B) $\frac{7}{3}$
(C) $\frac{6}{3}$ (D) $\frac{2}{3}$

Q2. The improper fraction **$\frac{11}{4}$** as a mixed number is:

- (A) 2 and $\frac{1}{4}$ (B) 2 and $\frac{3}{4}$
(C) 3 and $\frac{1}{4}$ (D) 1 and $\frac{3}{4}$

Q3. What is **1 and $\frac{1}{2}$ + 1 and $\frac{1}{2}$** ?

- (A) 2 (B) 2 and $\frac{1}{2}$
(C) 3 (D) 3 and $\frac{1}{2}$

Q4. Which is GREATER: **$\frac{7}{4}$ or 1 and $\frac{1}{2}$** ?

- (A) $\frac{7}{4}$ (B) 1 and $\frac{1}{2}$
(C) Both equal (D) Cannot compare

Q5. What is **$\frac{4}{9} + \frac{2}{9} + \frac{1}{9}$** ?

- (A) $\frac{7}{27}$ (B) $\frac{7}{9}$
(C) $\frac{6}{9}$ (D) $\frac{8}{9}$

Q6. A pizza is cut into 8 slices. Tara eats 3 slices and Veer eats 2 slices. What FRACTION of the pizza remains?

- (A) $\frac{5}{8}$ (B) $\frac{3}{8}$
(C) $\frac{2}{8}$ (D) $\frac{1}{2}$

Q7. What is **$2 - \frac{3}{4}$** ?

- (A) 1 and $\frac{1}{4}$ (B) 1 and $\frac{1}{2}$
(C) 1 and $\frac{3}{4}$ (D) $\frac{3}{4}$

Q8. The simplest form of **$\frac{12}{18}$** is:

- (A) $\frac{6}{9}$ (B) $\frac{4}{6}$
(C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Q9. Asha drinks **$\frac{1}{4}$ L** of milk every day. How much milk does she drink in ONE WEEK?

- (A) 1 L (B) 1 and $\frac{1}{4}$ L
(C) 1 and $\frac{3}{4}$ L (D) 2 L

Q10. Which fraction is equal to **3** (three wholes)?

- (A) $\frac{3}{3}$ (B) $\frac{9}{3}$
(C) $\frac{3}{9}$ (D) $\frac{6}{3}$



Q11. What is $\frac{5}{12} + \frac{7}{12}$?

- (A) $\frac{12}{24}$ (B) 1
(C) 2 (D) $\frac{11}{12}$

Q12. A rope is **2 and $\frac{1}{2}$ m** long. Another rope is **1 and $\frac{3}{4}$ m**. What is their TOTAL length?

- (A) 3 and $\frac{1}{4}$ m (B) 4 and $\frac{1}{4}$ m
(C) 4 m (D) 3 and $\frac{3}{4}$ m

Q13. Which point on a number line lies between 1 and 2?

- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$
(C) $\frac{5}{2}$ (D) $\frac{2}{3}$

Q14. Mohan did $\frac{2}{3}$ of his 27 homework sums before dinner. How many sums are LEFT for after dinner?

- (A) 18 (B) 12
(C) 9 (D) 6

Q15. Which of these is EQUAL to $\frac{5}{4}$?

- (A) 1 and $\frac{1}{4}$ (B) 1 and $\frac{1}{2}$
(C) 1 and $\frac{3}{4}$ (D) 2 and $\frac{1}{4}$

Q16. A cake recipe needs $\frac{3}{4}$ cup of milk. Nina is making HALF the recipe. How much milk does she need?

- (A) $\frac{1}{4}$ cup (B) $\frac{3}{8}$ cup
(C) $\frac{1}{2}$ cup (D) 1 and $\frac{1}{2}$ cups

Q17. What must be ADDED to $\frac{2}{5}$ to get **1 and $\frac{1}{5}$** ?

- (A) $\frac{3}{5}$ (B) $\frac{4}{5}$
(C) $\frac{1}{5}$ (D) $\frac{2}{5}$

Q18. Out of 60 marbles, Raju gives $\frac{1}{5}$ to his sister and $\frac{1}{4}$ to his friend. How many marbles does Raju KEEP?

- (A) 27 (B) 33
(C) 30 (D) 36

Q19. How many **quarters ($\frac{1}{4}$)** make up **2 and $\frac{3}{4}$** ?

- (A) 9 (B) 10
(C) 11 (D) 12

Q20. A water bottle holds 1 L. Sara fills $\frac{3}{4}$ of it, drinks $\frac{1}{2}$ of the bottle's capacity, then adds $\frac{1}{4}$ more. How full is the bottle now?

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) $\frac{3}{4}$ (D) Full



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	B	C	A	B	B	A	C	C	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	B	B	C	A	B	B	B	C	B

Step-by-Step Solutions

Q1. Correct answer: (B)

2 wholes = $6/3$; plus $1/3 = 7/3$. (Multiply whole $\times$ denominator, add numerator: $2 \times 3 + 1 = 7$.)

Q2. Correct answer: (B)

$11 \div 4 = 2$ remainder 3, so $11/4 = 2$ and $3/4$.

Q3. Correct answer: (C)

$1\frac{1}{2} + 1\frac{1}{2} = 2$ wholes + 2 halves = $2 + 1 = 3$.

Q4. Correct answer: (A)

$7/4 = 1\frac{3}{4}$ and $1\frac{1}{2} = 1\frac{2}{4}$. Since $1\frac{3}{4} > 1\frac{1}{2}$, $7/4$ is greater.

Q5. Correct answer: (B)

Same denominators: $4 + 2 + 1 = 7 \rightarrow 7/9$.

Q6. Correct answer: (B)

Eaten = $3 + 2 = 5$ slices. Remaining = $8 - 5 = 3$ slices $\rightarrow 3/8$.

Q7. Correct answer: (A)

$2 = 8/4$. $8/4 - 3/4 = 5/4 = 1$ and $1/4$.

Q8. Correct answer: (C)

Divide top and bottom by 6: $12 \div 6 / 18 \div 6 = 2/3$. ($6/9$ and $4/6$ are equal but not simplest.)

Q9. Correct answer: (C)

$7 \times 1/4 = 7/4 = 1$ and $3/4$ L.

Q10. Correct answer: (B)

$9/3 = 9 \div 3 = 3$. ($3/3 = 1$ and $6/3 = 2$.)

Q11. Correct answer: (B)

$5 + 7 = 12$, so the sum is $12/12 = 1$ whole.

Q12. Correct answer: (B)

$2\frac{1}{2} = 10/4$ and $1\frac{3}{4} = 7/4$. Sum = $17/4 = 4$ and $1/4$ m.

Q13. Correct answer: (B)

$3/2 = 1\frac{1}{2}$, which lies between 1 and 2. ($1/2$ and $2/3$ are below 1; $5/2 = 2\frac{1}{2}$ is beyond 2.)

Q14. Correct answer: (C)

Done = $2/3$ of 27 = 18. Left = $27 - 18 = 9$ sums.

Q15. Correct answer: (A)

$5 \div 4 = 1$ remainder 1, so $5/4 = 1$ and $1/4$.

**Q16. Correct answer: (B)**

Half of $\frac{3}{4} = \frac{3}{8}$ cup. (Halving a fraction doubles the denominator: $\frac{3}{4} \rightarrow \frac{3}{8}$.)

Q17. Correct answer: (B)

1 and $\frac{1}{5} = \frac{6}{5}$. Needed = $\frac{6}{5} - \frac{2}{5} = \frac{4}{5}$.

Q18. Correct answer: (B)

Sister = $60 \div 5 = 12$; friend = $60 \div 4 = 15$. Given away = 27. Kept = $60 - 27 = 33$.

Q19. Correct answer: (C)

$2\frac{3}{4} = \frac{11}{4} \rightarrow 11$ quarters.

Q20. Correct answer: (B)

Start $\frac{3}{4} \rightarrow$ drink $\frac{1}{2}$: $\frac{3}{4} - \frac{2}{4} = \frac{1}{4} \rightarrow$ add $\frac{1}{4}$: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ full.

